



#	Milestone	Generative AI Guidelines	Due	Mark Release
3	Proposed Solution	D: AI Task Completion, Human Evaluation ¹	Nov. 4, Wed., AoE ²	Nov. 9, Mon.

This course is *project driven*. Students will work in teams to define, develop, and complete a project in the field of data mining/machine learning over the course of one semester. Each team will consist of undergraduate students and one graduate student, who serves as the project lead, coordinating the project and guide the team through the research and development process. The course project has two main objectives:

- Hands-on experience: Gain practical experience in applying data mining and machine learning methods to a real-world problem, while developing skills in project planning, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating a solution.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful research while completing the course requirements.

A research project typically starts with a problem and proposes a solution by designing, implementing, and evaluating an algorithm or computational approach. In this sense, it may resemble a software project; however, there are important differences. A software project may primarily focus on implementing or integrating existing algorithms to build a functional system. In contrast, a research project should aim to investigate a problem, identify a limitation or gap in existing approaches, and propose a solution that addresses that gap. The proposed solution may involve (1) a new algorithm, (2) a modification or extension of an existing algorithm, (3) a new combination/pipeline of existing techniques, or (4) a novel application of existing methods to a problem where they have not been adequately explored. Therefore, the goal is not simply to implement an existing method, but to make a meaningful contribution beyond what has already been done, supported by appropriate experiments and evaluation. Roughly, a research project consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)
3. **Solution Design & Implementation (Proposed Solution or Method)**
4. Evaluation (Experimentation)
5. Project Presentation: 3-Minute Talk (3MT)
6. Project Management & Documentation (Graduate Student Only, Umbrella Activity)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 3: Solution Design & Implementation, also known as Proposed Solution or Method.

3.1. Formal Definition: Input & Output (Solution)

At a very high level of abstraction, any proposed method for solving a problem can be viewed as a *function* that takes some input, performs a finite sequence of processing and/or calculations, and produces an output (i.e., a solution). For example, suppose your proposed method is to “*cluster people into groups of like-minded people*”.

¹ AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI-generated content and evaluating its output (e.g. critique or evaluation of AI-generated analysis). You will use AI to complete specific tasks in your assessment. Any AI-created content must be cited.

² <https://www.timeanddate.com/time/zones/aoe>



Informally, the input is “a collection of people”, and the output is “a collection of groups in which each person belongs to a group”. This informal description, however, is *not* precise. Different readers may interpret it differently:

- Can a person belong to *multiple* groups?
- What exactly does *like-minded* mean?
- What does a *collection* mean? Is it a set, list, stack, ...?
- What *constraints* must the resulting groups satisfy?
- How do we determine whether two people are *similar*?

To ensure that all readers have the same interpretation, we use the language of logic and mathematics to formally define our proposed solution. We already performed a similar step when formally defining the problem in the previous Milestone 1: Project Definition & Team Formation. For example, suppose we define the input as “a set of people P ”. Our proposed solution is “a function f that takes P as input and returns a partition of P . A partition consists of non-empty subsets of P such that every person belongs to exactly one and only one subset of P .”

“Given a set of people $P = \{p_i\}_{i=1}^N$, our proposed solution is a function f such that $f(P)$ outputs a partition on P that includes subsets of P such that each person $p_i \in P$ is a member of one and only one subset (group), and the union of all subsets (groups) is equal to P . Formally,

$$f: P \rightarrow \text{Powerset}(P) \\ f(P) = \{C_j \mid C_j \subseteq P, |C_j| > 0, \forall C_j, C_{k \neq j} C_i \cap C_j = \emptyset\}$$

where Powerset of P is the set of all subsets of P . As an example, if $P = \{p_1, p_2, p_3\}$, then $f(P) = \{\{p_1, p_3\}, \{p_2\}\}$.”

Notice that we have not yet explained how f produces this output. We have only formally specified what the input is and what constitutes a valid output.

3.2. Formal Definition: Algorithm

So far, we have formally specified the input and output of our proposed solution. We now need to describe how our method transforms the input into the desired output. In a Data Mining or Machine Learning project, this usually means specifying how the raw data are:

- represented in a form suitable for computation;
- preprocessed or transformed;
- used to construct relevant embeddings, points, features or representations;
- provided to a data mining or machine learning algorithm/model; and
- transformed into the final prediction, ranking, clustering, recommendation, or other solution.

For example, to “cluster people into groups of like-minded people”, we first need to convert the raw information about people into a representation that a machine learning algorithm can process.

“In our work, we represent each person $p_i \in P$ by the documents that she has published within a time period T . So, given the set of all documents $D = \{d_{i,t}\}$, published by all persons in P , where $d_{i,t}$ is the document published by the person p_i at time t ; $1 \leq t \leq T$, we calculate the similarity of a pair of person (p_i, p_j) by calculating the similarity of their documents as follows:

$$g(p_i, p_j) = \sum_{t=1}^T h(d_{i,t}, d_{j,t})$$

where h is a function that calculates the similarity of a pair of documents based on the number of words that are shared, i.e., the Jaccard distance:

$$h(d_1, d_2) = \frac{d_1 \cap d_2}{d_1 \cup d_2}$$

where d_i is a set of words in the document. Finally, we apply a portioning clustering technique using g as the similarity (distance) measure to produce the subsets of people.”

This is only one possible definition of proposed solution. For a different project, the components of this pipeline may be different. Your proposed solution should be sufficiently formal that a knowledgeable reader can understand exactly what your method does, without having to guess important details. Depending on your project, this may include:

- distance or similarity functions + justify the chosen function;
- model architecture or algorithm + justify the chose model;
- important parameters and hyperparameters;
- training procedure, if applicable, + justification;
- prediction/inference procedure + justification;

You do not need to mathematically formalize every line of your implementation. Instead, focus on the key computational steps that define your proposed method.

Important: A machine learning algorithm is *not* simply saying “we use k -means,” “we use a neural network,” or “we use a Random Forest.” Your proposed solution should explain how the data are represented, what information is provided to the model, what the model computes, and how its output becomes the final solution.

Important: Formalizing a proposed solution can be challenging. That is expected. The goal is *not* to make the mathematics unnecessarily complicated, but to remove ambiguity and precisely communicate what your proposed method is supposed to produce.

3.3. Formal Definition: Pseudo-code

You need to support your formal algorithmic description with pseudo-code. Pseudo-code is *not* your program written line-by-line in a specific programming language. Instead, it describes the overall logic and flow of your solution using a language-independent notation, including important steps necessary to transform the input into the final solution, such that a reader can understand what the solution does without reading the source code.

Algorithm 2 Finding regions of like-mindedness for time interval t ($\mathcal{R}_t$)

Inputs:
 c , homogeneity condition;
 G_t , multigraph at time interval t ;
 U , set of users;
 Z , set of topics of interest;

Output:
 $\mathcal{R}_t$, set of regions of like-mindedness for time interval t

Initialization:
 $\mathcal{R}_t = \emptyset$;
 $\text{find_r.t}(r = U \times \emptyset, C = [z_1, z_1, z_2, \dots, z_{|Z|}, z_{|Z|}])$;

```

1: procedure find_r.t( $r = A \times B, C$ )
2:   if ( $r \models c$ )  $\wedge$  ( $\exists r' \in \mathcal{R}_t : r \subset r'$ ) then
3:      $\forall r'' \in \mathcal{R}_t$  if  $r'' \subset r$  then  $\mathcal{R}_t \leftarrow \mathcal{R}_t \setminus r''$ 
4:      $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup r$ 
5:   for all  $z_j \in Z$  do
6:      $A \leftarrow r.A; B \leftarrow r.B \cup z_j; C \leftarrow C \setminus z_j$ 
7:     if  $r.B = \emptyset$  then find_r.t( $A \times B, C$ )
8:   else
9:     for all  $z_i \in r.B$  do
10:      for all  $(z_i \rightarrow z_j) \in \mathcal{U}_t$  do
11:         $A \leftarrow r.A \cap \mathcal{U}_{z_i z_j, t}$ 
12:        find_r.t( $A \times B, C$ )

```

³ The FAIR Guiding Principles for scientific data management and stewardship: <https://www.nature.com/articles/sdata201618>

3.5.1. Reproducible and Repeatable Results

Your implementation should also produce reproducible/repeatable results when the same input and settings are used. There are different definitions and distinctions between reproducibility and replicability in the academic community. For the purpose of this course, the practical expectation is that *“given the same input data, code, configuration, and random seed, another person should be able to run your implementation and obtain the same results.”* For instance, if your report says that if $P=\{p_1, p_2, p_3\}$, then $f(P)=\{\{p_1, p_3\}, \{p_2\}\}$, then someone downloading your repository should be able to provide the same input $P=\{p_1, p_2, p_3\}$ and obtain the same result $f(P)=\{\{p_1, p_3\}, \{p_2\}\}$ under the same experimental settings. Many machine learning algorithms contain randomized operations. In such cases, you should explicitly control the random seed where possible. For example, in Python:

```
import random
random.seed(0)
import torch
torch.manual_seed(0)
import numpy as np
np.random.seed(0)
```

If your method uses additional libraries or sources of randomness, you should also consider whether their random states need to be controlled.

3.6. Submission Guidelines

2.2.1. Deliverables

- Submission must be written in English, in the current ACM two-column conference format in LaTeX. Overleaf⁴ templates are available from the ACM Website⁵. Use the *sigconf* proceedings template.
- The submission must be *exactly 2 pages* in length, no more and no less, including author list, figures, tables, and references, and must be authored by all team members, with the graduate student listed last in the author list. We understand that fitting all the required content into exactly two pages can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially writing a longer version (e.g., 5-10 pages) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required two-page limit.
- The submission must be a single zip file named: 3_Solution_Firstname1_Firstname2_[...].zip, e.g., 3_Solution_Reia_Ali_Luiz_Hossein.zip, containing:
 - o The LaTeX source files and any required supplementary files;
 - o The compiled two-page PDF file.

A *sample* is attached to this manual and is also available [here](#)⁶ (2_Solution.tex).

2.2.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...) ⁷ for the project and linked to

⁴ <https://www.overleaf.com/gallery/tagged/acm-official>

⁵ <https://www.acm.org/publications/proceedings-template>

⁶ <https://www.overleaf.com/read/bsrxtwqgggsb>

⁷ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.

the Abstract. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The lab instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.

- The graduate student serves as the project lead, responsible for coordinating the process, maintaining the overall project direction, and ensuring that the project develops toward a meaningful contribution. However, the project lead is also expected to be an active contributor, participating in activities such as problem formulation, literature review, methodology development, implementation, experimentation, analysis, and writing.
- Undergraduate members contribute to the project activities under the guidance of the project lead. All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.
- As part of **Milestone 6. Project Management & Documentation (Graduate Student Only)**, the graduate student (project lead) is expected to document each member's contributions throughout the project, with a brief contribution statement provided at each major project milestone. These statements will be compiled into the final project management report.

2.2.3. Marking Scheme

- 05% Author list, style, format, repo, file naming convention, ...
- 05% Abstract + Introduction
- 20% Formal Definition
- 10% Pseudo-code
- 10% Illustrative Figure
- 50% Code (FAIR-compliant) Please also see Appendix A: Recommended Code Structure.
 - o 20% Architecture (Layers, driver, ...)
 - o 10% README (Project bio, installation, features, license, ...)
 - o 10% Quickstart script
 - o 10% Reproducible/Repeatable results

Commented [A2]: Making the repo public is at your team's discretion. We do not require public repo for this course due to students' personal information and privacy, but it should be accessible to instructors and markers for evaluation purposes.

Appendix A: Recommended Codebase Structure

This is a template suggested for project implementation based on my experience in data science, machine learning, and software development projects. It is important to note that

- This is a general template and should be adjusted for any projects' need,
- The main goal is to make your codebase useful enough that other researchers can reproduce and replicate your work, fork your repository, build on top of it, and happily choose it as a baseline; without needing your involvement! The more people who use, extend, and cite your code, the more visible and impactful your work becomes.
- Your suggestions are more than welcome.

README.md

1. Intro [Overview]
2. Illustrative Sample
3. Prerequisite [Package Requirements]
4. Installation Guide
5. Quick Start



6. Results
7. Citation
8. License

requirements.txt *#for dependencies of this project and other libs and packages*

environment.yml *#for creating conda environment with automatic installation of required packages*

\data

- \raw *#the original dataset for the testbed*
- \preprocessed *#the final dataset required for this project*

\output *#the output of the codes in src such as models, logs, ...*

\evaluation

- \ground_truth [gold_standard]
- \results *#comparison results between the proposed method and the baselines*
 - o charts
 - o flat files
- \reports:

\src

- \libs *#any opensource library that are customized for this project*
- \cmn *#any common code to other parts of the project*
- \[method] *#the implementation of the proposed method with a proper name*
- main.[py, c, ...] *#the main entry of the project's pipeline*